

Research Proposal

Title: Machine Learning-Driven Predictive and Cost-Aware Autoscaling Framework for Kubernetes-Based Cloud-Native Applications

Introduction

Cloud-native applications built on microservices and Kubernetes have become central to modern software engineering. Kubernetes primarily relies on reactive autoscaling, which responds after resource utilisation exceeds predefined thresholds. This behaviour can increase response time during workload spikes and lead to unnecessary cloud costs after demand decreases. This research proposes a machine learning-driven predictive and cost-aware autoscaling framework to address these limitations.

Research Problem

Reactive autoscaling cannot anticipate workload changes. Consequently, applications may experience temporary performance degradation, while cloud resources remain over-provisioned after traffic subsides. There is a need for an intelligent framework that predicts future workload demand and balances performance with infrastructure cost.

Aim and Objectives

Aim: Design, implement and evaluate a machine learning-driven predictive and cost-aware autoscaling framework for Kubernetes-managed cloud-native applications.

Objectives:

- Analyse limitations of Kubernetes HPA.
- Develop workload prediction models.
- Implement proactive scaling and intelligent descaling.
- Integrate cost-aware scaling decisions.
- Compare results with Kubernetes HPA.

Methodology

A Kubernetes-based microservices application will be developed and tested on Minikube, followed by validation on Amazon EKS. Historical metrics collected through Prometheus will be used to train Random Forest, XGBoost and LSTM models. The best-performing model will drive predictive scaling decisions. Performance will be evaluated using response time, throughput, CPU and memory utilisation, scaling latency, availability and estimated cloud cost.

Expected Contributions

The study will contribute a machine learning-based workload prediction model, a cost-aware predictive autoscaling framework, an intelligent descaling mechanism, and an

experimental comparison against Kubernetes' Horizontal Pod Autoscaler under realistic cloud-native workloads.

Expected Outcomes

The framework is expected to improve application performance, optimise resource utilisation and reduce cloud infrastructure costs while maintaining service availability. The research will provide practical recommendations for intelligent resource management in Kubernetes-based cloud-native systems.

Keywords

Cloud Computing; Kubernetes; Cloud-Native Applications; Predictive Autoscaling; Machine Learning; Cost Optimisation; DevOps; AWS EKS.

Selected References

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